

Warren Liu

EDUCATION

University of Waterloo

Bachelor in Biomedical Engineering (co-op) | 4-month co-op terms

2025 - 2030

GPA: 3.87/4

SKILLS

Programming: C++, Python, MATLAB, HTML, AI coding agents (Codex, Claude Code, Kimi)

Software: SolidWorks (Part, Assembly, Drawings), GitHub, Git, DBeaver, Docker, CloudFlare R2, Microsoft Suite

Tools: Microscopes, Digital Force Gauge, Digital Multimeter, 3D printer, Adobe Illustrator (Laser cutter designs)

Certifications: WHIMS 2015, AI Fluency: Framework & Foundations

TCPS 2: CORE-2022, MATLAB Onramp, AODA

Languages: Fluent in English and Mandarin, working proficiency in French

EXPERIENCE

Software Engineering Intern – Startup GRC Platform (SureStep)

May - Aug. 2026

- Engineered and maintained a fullstack data pipeline processing 10M+ records, supporting large-scale regulatory data ingestion and database integration.
- Developed API feeds and data integrations to ingest external regulatory information into the company's GRC platform.
- Built automated Python workflows to scrape, parse, normalize, and convert Canadian and U.S. state regulations into structured JSON, enabling ingestion into production databases.
- Utilized OpenAI Codex and Claude Code efficiently to prototype mockups, implement and refine over 50 React and TypeScript frontend features, improving the UI/UX of the GRC platform.
- Communicated technical functionality and implementation details to coworkers, translating complex software concepts into clear and intuitive explanations.

Board Game Accessibility Project

Sept. - Dec. 2025

- Conducted user-centered design process (empathy interviews, ideation, rapid prototyping, testing) to redesign Battleship for users with manual impairments.
- Performed quantitative evaluation of prototypes using Excel analysis.
- Developed C++ program to automate normalization and metric weighing for QFD table; reducing evaluation time by over 50%.

PROJECTS

Portfolio Website Project

Feb. 2026 - Present

- Developed a multi-page portfolio website using HTML and CSS, working with VSCode and GitHub to present engineering projects, technical skills, and creative works.

3D-Printing Projects

Sept. 2025 - Dec. 2025

- Designed a multi-component 3D puzzle in a team setting, performing tolerance and error calculations to ensure accurate 3D printing using SolidWorks Assembly and Z-Suite.
- Recreated 3D-print-ready CAD models of fictional vehicles and generated accurate ANSI-compliant orthographic drawings and BOMs in SolidWorks.

COMMUNITY INVOLVEMENTS

Sept. 2022 - June 2025

- Managed a 40+ member STEM club, coordinating event logistics and guest presentations while providing younger students with insights into upper-grade science courses and university application processes.
- Assisted 30+ students for 6 hours weekly with coursework across various grades through a high school peer tutoring program.
- Taught English to students in different parts of China through online video conferencing platforms with the Read Speak Lead Association for over 40 hours.